FARMCONNECT

Testing & ML Performance Results | Phase III Evidence

Based on the FarmConnect Capstone Report (Mid-Term), academic session 2026-27. This document consolidates reported project observations; it does not claim independent reproduction or additional benchmark runs.

1. Crop Disease Classification Experiment

The documented implementation uses MobileNetV2 transfer learning with TensorFlow/Keras on a PlantVillage-derived dataset of 38 crop/healthy/disease classes. A saved .keras model is used by the Python inference script and Express prediction endpoint.

Reported metric	Recorded result
Held-out test images	8,146
Accuracy	94.88%
Macro precision	94.13%
Macro recall	92.93%
Macro F1-score	93.17%

Interpretation: These are aggregate recorded test results. An individual prediction confidence score is not the same as overall test accuracy. Independent field-image validation, split-leakage checks and confidence calibration remain pending.

2. API and Functional Test Observations

Component / test	Recorded observation
Official mandi data API	Corrected data.gov.in resource returned HTTP 200 with published records; an earlier resource returned HTTP 500 during development.
Mandi module	Authenticated lookup, crop/state/district/optional market filters, cache and user-visible unavailable-state handling implemented.
Authentication and dashboard	Local farmer sign-in and dashboard demonstration documented.
Profile and Firestore	Profile fields and PATCH allowlist updated; save route implemented. Full regression testing advised.
State / district	Linked local JSON selection; Bihar district options observed in browser.
Photo picker	Local selection and preview; browser-only storage.
Scheme hub	Model/routes and verified-only display policy exist; verified published content pending.
JavaScript checks	node --check used during frontend/backend patches.

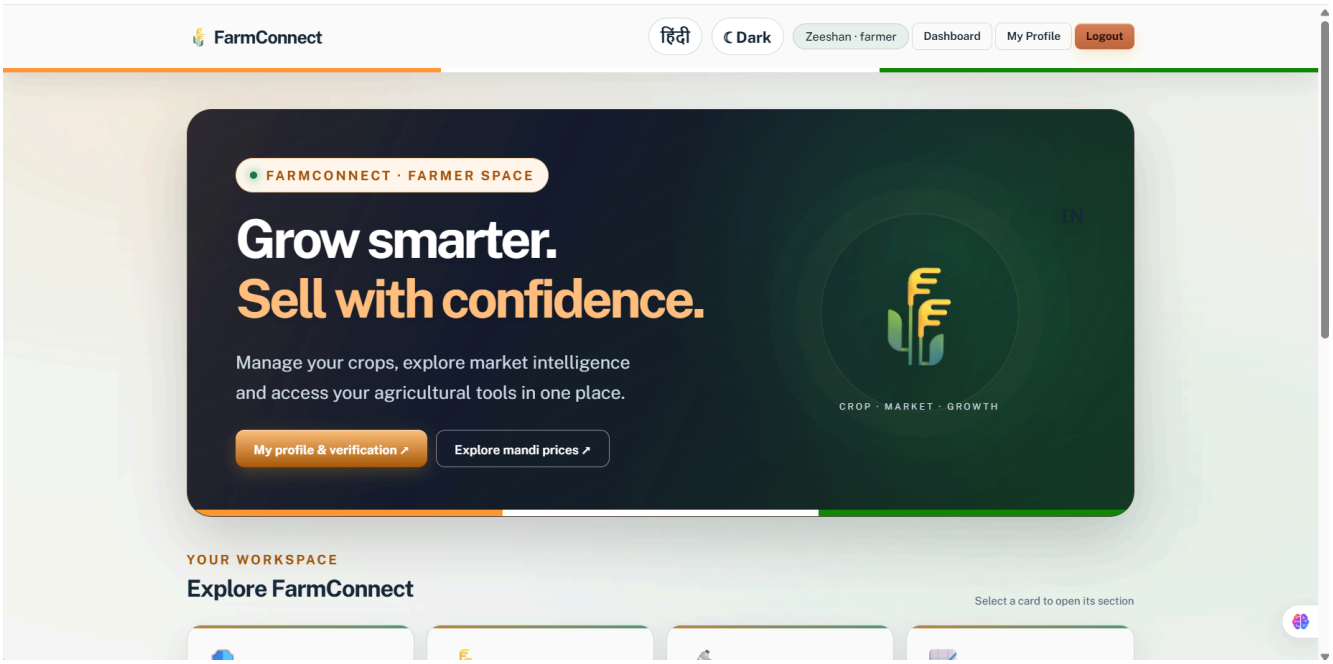
3. Validation Boundaries

No measured API latency, throughput, error rate, deployment uptime or hardware benchmarks are documented. Successful HTTP 200 retrieval does not guarantee same-day prices or continuous availability. Authorized identity

verification, production payments and complete marketplace transactions are not validated.

4. Prototype Screenshot Evidence

Farmer dashboard screenshot reproduced from the supplied project evidence. It illustrates the interface, not production readiness of every linked feature.



Source: FarmConnect Capstone Report (Mid-Term), sections 7-12, Figure 6. Original test logs, confusion matrix, dataset version and demonstration recording should be attached separately when available.